

■ Agentic AI, MCP Server ■■■ RAG

■■■■■■■■■■ Learning Roadmap — Bengali Guide

Background: Python Basic + RHCE (RHEL 6.0) | Target: AI Engineering

■ PHASE OVERVIEW — ■■■■■■ ■■ ■■■■■■ ■■■■■■?

Phase	■■■■■	■■■■
Phase 0	Python Refresh + Essential Tools	■-■ ■■■■■■
Phase 1	LLM ■ GenAI Fundamentals	■ ■■■■■■
Phase 2	RAG (Retrieval-Augmented Generation)	■-■ ■■■■■■
Phase 3	Agentic AI (LangChain → LangGraph → CrewAI)	■-■ ■■■■■■
Phase 4	MCP Server (Model Context Protocol)	■-■ ■■■■■■
Phase 5	Real-World Projects + Deployment	■-■ ■■■■■■

■■■ ■■ Order? — RAG ■■■■■■ ■■■■ Agent-■■■ 'memory' ■■■■■■ Agent ■■■■■■ ■■■■
MCP-■■■ ■■■■■■■■ ■■■■■■■■ ■■ sequence follow ■■ ■■■■■■ MCP ■■■■■■ ■■■■■■ confused
■■■■■ ■■■■ ■■■■: ■-■ ■■■■ (■■■■■ ■-■ ■■■■■■ ■■■■■)

■ PHASE 0 — Python Refresh + Essential Tools

AI engineering-■ ■■■■■■■■ ■■■■ ■■■■ ■■ Python features:

- async/await (asyncio) — MCP server-■■■ ■■■■ ■■■■■■■■
- Decorators — FastMCP-■ @mcp.tool() decorator ■■■■■■■■ ■■■■
- Type hints + Pydantic — Agent-■■■ structured output-■■■ ■■■■
- Virtual Environment (venv / uv) — project isolation
- API calls (requests, httpx) — external tools connect ■■■■

YouTube Resources

■ Python Async IO Complete Guide English <https://www.youtube.com/watch?v=t3JUfblc2kc>

■ Krish Naik — Python for GenAI
(Hindi/English) Hindi <https://www.youtube.com/watch?v=7qqGnuRrWxg>

Website / Documentation

■ Learn RAG From Scratch — LangChain Engineer (freeCodeCamp)	English	https://www.youtube.com/watch?v=sVcwVQRHlc8
■ Complete RAG Crash Course — Krish Naik (■ ■ ■ ■ ■ ■ ■ ■)	Hindi/En	https://www.youtube.com/watch?v=o126p1QN_RI
■ RAG with LangChain Full Tutorial	English	https://www.youtube.com/watch?v=YLPNA1j7kmQ
■ Production RAG with LangChain & Vector Databases	English	https://www.youtube.com/watch?v=mHxLXzYjQRE

■ Website / Blog / Course

■ LangChain Official RAG Tutorial	https://python.langchain.com/docs/tutorials/rag/
■ RAG from Scratch — LangChain GitHub	https://github.com/langchain-ai/rag-from-scratch
■ DataCamp — RAG with LangChain Course	https://www.datacamp.com/courses/retrieval-augmented-generation-rag-wi.. ..
■ Advanced RAG Techniques — Towards Data Science	https://towardsdatascience.com/advanced-rag-techniques-an-illustrated-...

■ PHASE 3 — Agentic AI

■■■■■■ chatbot ■■■■ ■■■■■■ ■■■■■■ Agent ■■■■ plan ■■■■, tool ■■■■■■■■ ■■■■, ■■■■ steps ■■■■ decide ■■■■ User ■■■■ '■■■ research paper summary ■■■■ ■■■■ email ■■■■■■' — Agent ■■■■■■ PDF ■■■■■■, summary ■■■■■■■■, ■■■■ email send ■■■■■■

■■■■■■ ■■■■■■:

■■■■■	Framework	■■ ■■■■■■
Week 1–2	LangChain Basics	Chains, Prompts, Memory, Tool calling, Simple ReAct Agent
Week 3–4	LangGraph	State machines, Complex workflows, Human-in-the-loop
Week 5	CrewAI	Multi-agent systems, Role-based agents, Agent collaboration

■ YouTube Resources

■ Agentic AI Full Course — LangGraph Zero to Hero Part 1	English	https://www.youtube.com/watch?v=4RtKGasvNC0
■ Agentic AI Full Course — LangGraph Zero to Hero Part 2	English	https://www.youtube.com/watch?v=YIv_GDxJqbA
■ LangGraph Complete Course for Beginners	English	https://www.youtube.com/watch?v=jGg_1h0qzaM
■ Agentic AI Tutorial for Beginners — LangGraph	English	https://www.youtube.com/watch?v=CnXdddeZ4tQ
■ CrewAI Tutorial — Beginners Crash Course	English	https://www.youtube.com/watch?v=G42J2MSKyc8
■ Complete Agentic AI Course 10 Hours — Krish Naik	Hindi/En	https://www.youtube.com/watch?v=rV3HJ4LEZ7k

■ Website / Course

■ LangGraph Official Documentation	https://langchain-ai.github.io/langgraph/
■ CrewAI Official Documentation	https://docs.crewai.com
■ IBM Coursera — Agentic AI with LangGraph & CrewAI (Free Audit)	https://www.coursera.org/learn/agentic-ai-with-langgraph-crewai-autoge...
■ LangChain Official Learn Page	https://docs.langchain.com/oss/python/learn
■ Neural Maze — Agentic RAG Advanced Blog	https://theneuralmaze.substack.com

Agent Frameworks ■■■■■■:

Framework	■■■ ■■■■■■■■ ■■■■
LangChain	Basic chains, simple agents, ■■■■ ■■■■ ■■■■

LangGraph	Complex state-based workflows, production-grade
CrewAI	Multi-agent collaboration, role-based agents
AutoGen	Conversational multi-agent, research-oriented

■ PHASE 4 — MCP Server (Model Context Protocol)

MCP ■■■ Anthropic-■■ ■■■■ open protocol■ ■■■ ■■■■■ AI models (Claude, GPT) ■■■■■
■■■■■ tools, database, ■■■ APIs-■■ ■■■■ standardized ■■■■■ ■■■ ■■■■ ■■■■ ■■■
■■■■■■■ AI-■■ ■■■■ ■■■■■ integration ■■■■■ ■■■ — MCP ■■■■■ ■■■■ server
■■■■■■■ ■■ AI connect ■■■■ ■■■■■

MCP-■■■ ■■■■ Core Primitive:

Primitive	■■■■■
Tools	AI ■■ functions call ■■■■ ■■■■ (■■■■■: database query, email send)
Resources	File/data ■■ AI ■■■■■ ■■■■ (■■■■■: config files, documents)
Prompts	Pre-built prompt templates ■■ user ■■■■■■ ■■■■ ■■■■

■ YouTube Resources

■ Build MCP Server in 10 Minutes Python	English	https://www.youtube.com/watch?v=AE04ehbAE78
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■ Official Docs + Blog (■■■■■■■ Important)

■ MCP Official Documentation — Anthropic	https://modelcontextprotocol.io/docs/develop/build-server
■ Anthropic MCP Course — Free	https://anthropic.skilljar.com/introduction-to-model-context-protocol
■ MCP Python SDK — GitHub	https://github.com/modelcontextprotocol/python-sdk
■ Build Your First MCP Server — Towards Data Science	https://towardsdatascience.com/model-context-protocol-mcp-tutorial-bui...
■ Python MCP Server Guide — Real Python	https://realpython.com/python-mcp/
■ How to Build MCP Server — freeCodeCamp	https://www.freecodecamp.org/news/how-to-build-your-own-mcp-server-wit..

■■ PHASE 5 — Real-World Projects + Deployment

Project	■■■	■■■■■	Stack
Project 1	Personal Document Assistant (RAG)	■■■■■ PDF upload ■■■■, ■■■■■■ ■■■■■■■ ■■■■■ ■■■■	LangChain + ChromaDB + Streamlit

Project 2	Research Agent (Agentic AI)	Topic ■■■■ → Agent web search ■■■■ → Summary → Email	LangGraph + Tavily + Gmail Tool
Project 3	Custom MCP Server	■■■■■■ database-■■ ■■■■■ MCP Server, Claude Desktop-■ ■■■■■■■■	FastMCP + SQLite + Claude Desktop
Project 4	Multi-Agent System (Advanced)	Writer + Researcher + Editor Agent ■■■■■■■ ■■■■ ■■■■	CrewAI + LangGraph + RAG

■ Best Books — Reference

Tier 1 — ■■■■■■ ■■■■■■■■■■:

Book	Author	■■ ■■■■
Generative AI with LangChain (2nd Ed.)	Ben Auffarth, Leonid Kuligin	LangChain, LangGraph, RAG, Multi-agent — ■■ ■■■■■■ Production-ready■
AI Engineering	Chip Huyen	Production-level AI systems ■■■■■■ Deploy, scale, monitor■

■ Generative AI with LangChain — O'Reilly

<https://www.oreilly.com/library/view/generative-ai-with/9781837022014>

■ AI Engineering — Chip Huyen (Amazon)

<https://www.amazon.com/AI-Engineering-Building-Applications-Foundation..>

Tier 2 — Advanced:

Book	Author	■■ ■■■■
Build AI Agents with LangChain & LangGraph	Jude Max	Practical agent building, portfolio projects
Hands-On LLMs	Jay Alammar, Maarten Grootendorst	LLM internals, RAG, Agents — deep dive

■ Build AI Agents — Amazon

<https://www.amazon.com/Build-Agents-LangChain-LangGraph-Applications/d...>

■ Hands-On LLMs — Free Online Guide

<https://www.llm-course.com>

■ The Illustrated Transformer — Free (Jay Alammar)

<https://jalammar.github.io/illustrated-transformer/>

■■ Essential Tools & Stack

Category	Tool	Link
LLM Framework	LangChain / LangGraph	docs.langchain.com
Vector DB (Local)	ChromaDB	trychroma.com
Vector DB (Cloud)	Pinecone	pinecone.io
LLM Provider (Free)	Groq (fast + free API)	console.groq.com
LLM Provider	OpenAI	platform.openai.com
Local LLM	Ollama	ollama.ai
Multi-Agent	CrewAI	docs.crewai.com
MCP Server	FastMCP	github.com/jlowin/fastmcp
Monitoring	LangSmith	smith.langchain.com

